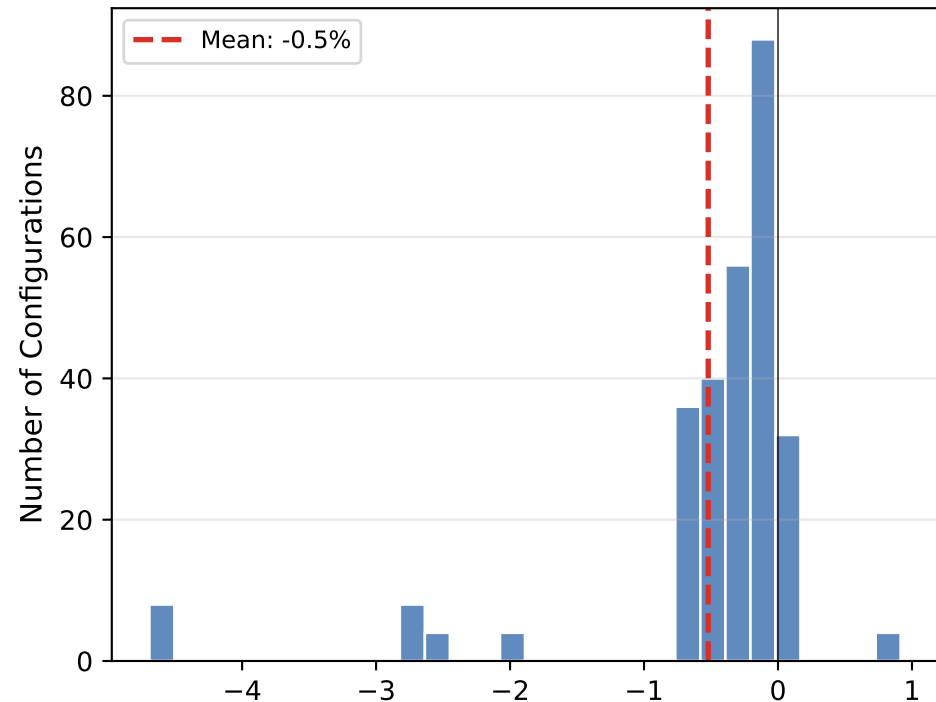


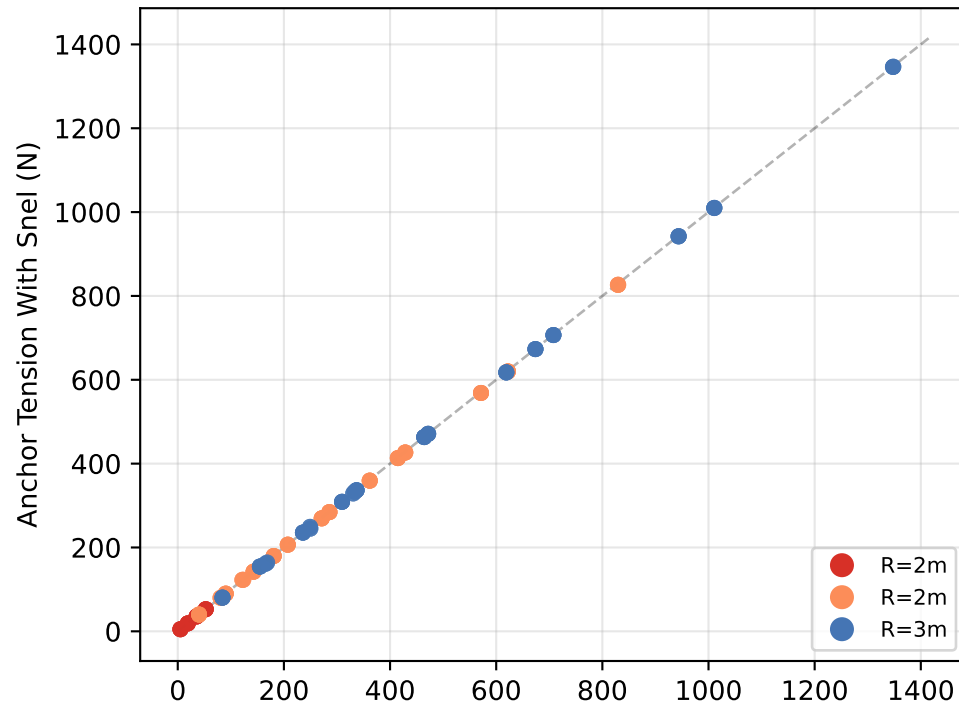
Snel Stall Delay: Does 3-D Correction Matter at System Level?

BEM v2.1 - 384-Config Sweep With and Without Snel
Snel Net Effect: All Viable Configurations



Thrust Change from Snel Stall Delay (%)

The Snel 3-D stall-delay correction has negligible net effect on anchor tension: mean change -0.5% (range -4.7% to 0.9%). 95% of configurations see less than 2.7% change in either direction. The correction boosts CL at the root (+32% at $r/R=0.1$) but this contributes little to total thrust because dynamic pressure is low at small radii. At design-point tip stations ($r/R>0.7$), the blade operates below stall and the correction is inactive. Snel matters for start-up and low-RPM conditions but does not affect steady-state autorotation thrust. The radial loading chart (bem-radial-loading) shows the full blade distribution. Data: 384-config BEM sweep, Snel on vs off, identical conditions.



Anchor Tension Without Snel (N)